How to Install Flutter on Windows?

Last Updated : 24 Jul, 2025

Flutter is Google's portable user interface (UI) toolkit. It is used to build and develop eye-catching, natively built mobile, desktop, and web applications from a single codebase. Flutter is free, open-sourced, and compatible with existing code. Due to its user-friendly interface and fairly simple yet to-the-point commands, it is utilized by companies and developers around the world.

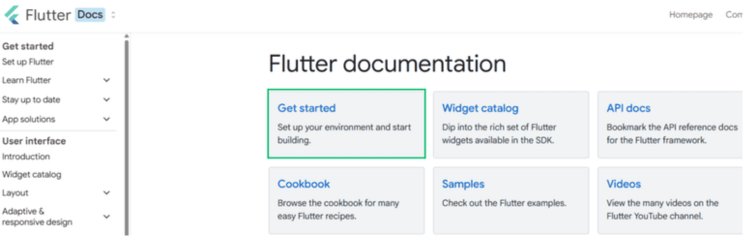
In this article, we will look into the process of installing Flutter on Windows.

Installing Flutter on Windows

Follow the below steps to install Flutter on Windows:

Step 1: Navigate Flutter in Webpage

Navigate to flutter.dev on your webpage. On the top menu bar, select **Docs > Get Started > Install > Windows** and then **choose your first type of app** as **Android**.



Flutter documentation

Get started
Set up your environment and start building.

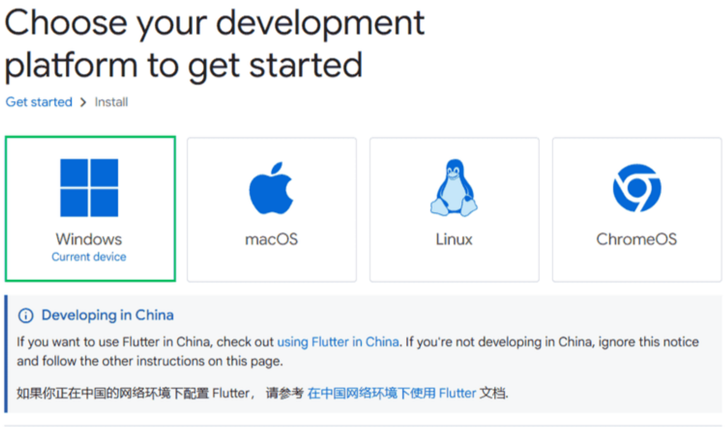
Widget catalog
Dip into the rich set of Flutter widgets available in the SDK.

API docs
Bookmark the API reference docs for the Flutter framework.

Cookbook
Browse the cookbook for many easy Flutter recipes.

Samples
Check out the Flutter examples.

Videos
View the many videos on the Flutter YouTube channel.

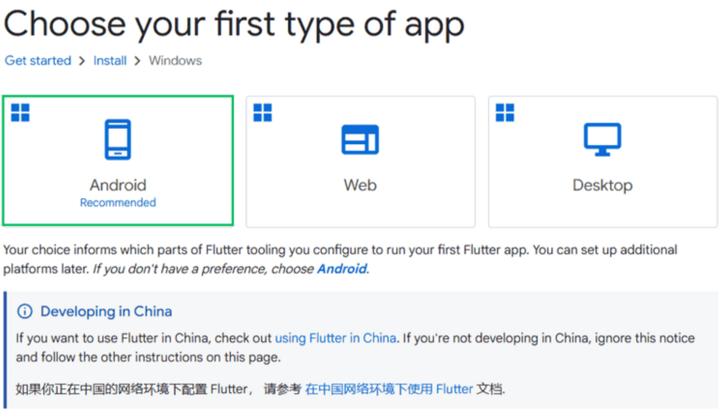


Choose your development platform to get started

Developing in China

If you want to use Flutter in China, check out using Flutter in China. If you're not developing in China, ignore this notice and follow the other instructions on this page.

如果你正在中国的网络环境下配置 Flutter， 请参考 在中国网络环境下使用 Flutter 文档.



Choose your first type of app

Get started > Install > Windows

Your choice informs which parts of Flutter tooling you configure to run your first Flutter app. You can set up additional platforms later. *If you don't have a preference, choose **Android**.*

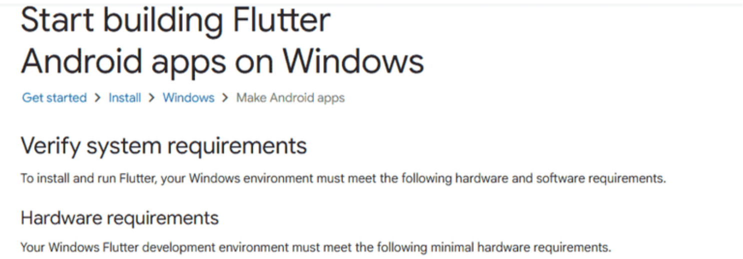
Developing in China

If you want to use Flutter in China, check out using Flutter in China. If you're not developing in China, ignore this notice and follow the other instructions on this page.

如果你正在中国的网络环境下配置 Flutter，请参考 在中国网络环境下使用 Flutter 文档.

Step 2: Check for the System Requirements.

Henceforth, you can begin the installation.



Start building Flutter Android apps on Windows

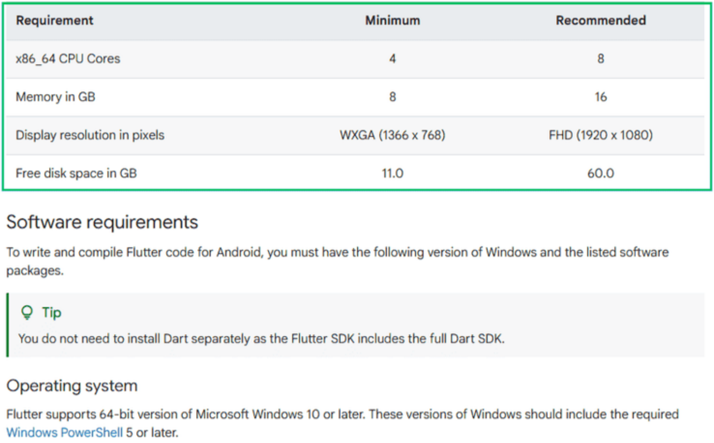
Get started > Install > Windows > Make Android apps

Verify system requirements

To install and run Flutter, your Windows environment must meet the following hardware and software requirements.

Hardware requirements

Your Windows Flutter development environment must meet the following minimal hardware requirements.



Requirement	Minimum	Recommended
x86_64 CPU Cores	4	8
Memory in GB	8	16
Display resolution in pixels	WXGA (1366 x 768)	FHD (1920 x 1080)
Free disk space in GB	11.0	60.0

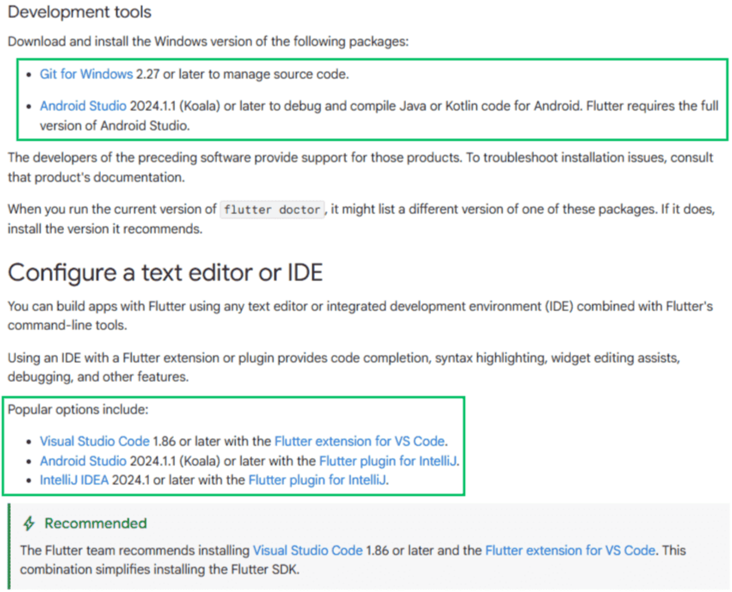
Software requirements

To write and compile Flutter code for Android, you must have the following version of Windows and the listed software packages.

> **Tip**
>
> You do not need to install Dart separately as the Flutter SDK includes the full Dart SDK.

Operating system

Flutter supports 64-bit version of Microsoft Windows 10 or later. These versions of Windows should include the required Windows PowerShell 5 or later.



Development tools

Download and install the Windows version of the following packages:

- Git for Windows 2.27 or later to manage source code.
- Android Studio 2024.1.1 (Koala) or later to debug and compile Java or Kotlin code for Android. Flutter requires the full version of Android Studio.

The developers of the preceding software provide support for those products. To troubleshoot installation issues, consult that product's documentation.

When you run the current version of `flutter doctor`, it might list a different version of one of these packages. If it does, install the version it recommends.

Configure a text editor or IDE

You can build apps with Flutter using any text editor or integrated development environment (IDE) combined with Flutter's command-line tools.

Using an IDE with a Flutter extension or plugin provides code completion, syntax highlighting, widget editing assists, debugging, and other features.

Popular options include:

- Visual Studio Code 1.86 or later with the Flutter extension for VS Code.
- Android Studio 2024.1.1 (Koala) or later with the Flutter plugin for IntelliJ.
- IntelliJ IDEA 2024.1 or later with the Flutter plugin for IntelliJ.

> **Recommended**
>
> The Flutter team recommends installing Visual Studio Code 1.86 or later and the Flutter extension for VS Code. This combination simplifies installing the Flutter SDK.

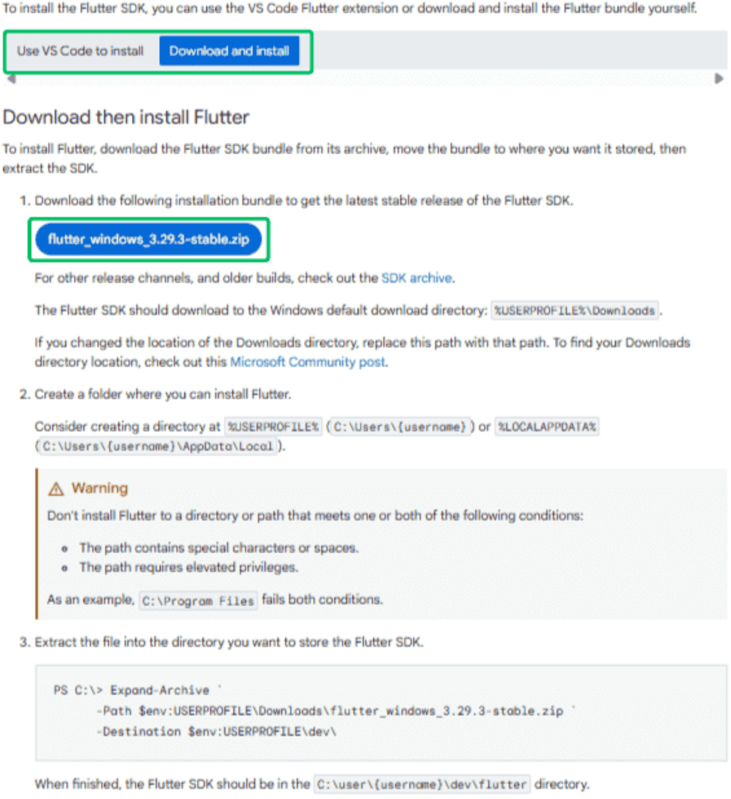
You can get a detailed procedure for installing the latest versions of Windows PowerShell 5.0 , Git for Windows, Visual Studio Code, Android Studio, IntelliJ IDEA if not already installed.

> *Note: But, It recommended to install Visual Studio Code.*

Step 3: Restart the system after installing Git and Visual Studio Code on your windows.

Once done, let's get to the installation of Flutter Software development Kit (Flutter SDK). We have two ways to Install the Flutter SDK as shown in the image below. Select **Download and install** and Click on the **"flutter_windows_version-stable.zip"** file.

Install the Flutter SDK



To install the Flutter SDK, you can use the VS Code Flutter extension or download and install the Flutter bundle yourself.

Download then install Flutter

To install Flutter, download the Flutter SDK bundle from its archive, move the bundle to where you want it stored, then extract the SDK.

1. Download the following installation bundle to get the latest stable release of the Flutter SDK.

   For other release channels, and older builds, check out the SDK archive.

   The Flutter SDK should download to the Windows default download directory: `%USERPROFILE%\Downloads`.

   If you changed the location of the Downloads directory, replace this path with that path. To find your Downloads directory location, check out this Microsoft Community post.

2. Create a folder where you can install Flutter.

   Consider creating a directory at `%USERPROFILE%` (`C:\Users\{username}`) or `%LOCALAPPDATA%` (`C:\Users\{username}\AppData\Local`).

   > ⚠ **Warning**
   >
   > Don't install Flutter to a directory or path that meets one or both of the following conditions:
   >
   > - The path contains special characters or spaces.
   > - The path requires elevated privileges.
   >
   > As an example, `C:\Program Files` fails both conditions.

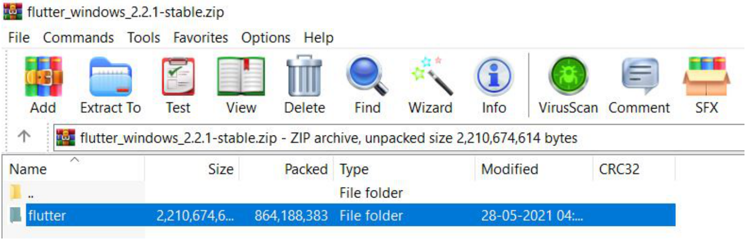
3. Extract the file into the directory you want to store the Flutter SDK.

```
PS C:\> Expand-Archive `
      -Path $env:USERPROFILE\Downloads\flutter_windows_3.29.3-stable.zip `
      -Destination $env:USERPROFILE\dev\
```

   When finished, the Flutter SDK should be in the `C:\user\{username}\dev\flutter` directory.

Flutter SDK is the tool that not only allows us to create flutter projects but also build those projects and transform them into native mobile applications. In simpler words, **Flutter SDK is the core tool for building a flutter UI.**

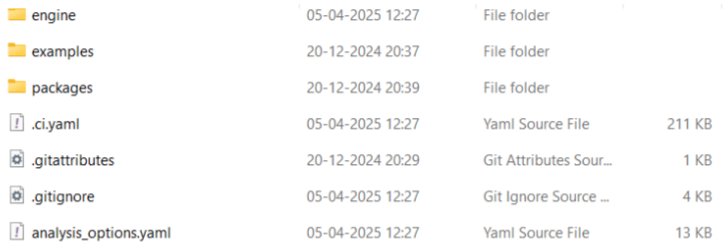
Once the zip file is downloaded, extract the **'flutter'** folder (drag and drop) to any path/directory of the system where you get the **read and write access**. Typically, it is better to create a new folder in a separate directory apart from the system drive due to permission issues (In my case, the target destination is **D: > development > flutter**).


flutter_windows_2.2.1-stable.zip
File Commands Tools Favorites Options Help
Add
Extract To
Test
View
Delete
Find
Wizard
Info
VirusScan
Comment
SFX

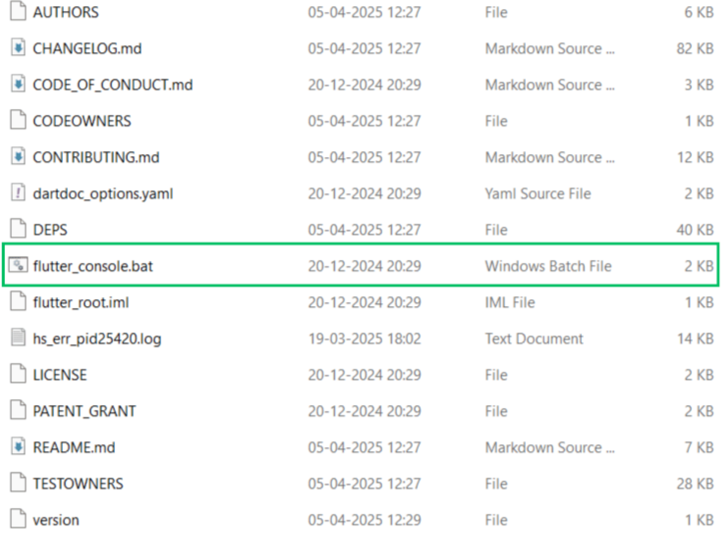
flutter_windows_2.2.1-stable.zip - ZIP archive, unpacked size 2,210,674,614 bytes

Name	Size	Packed	Type	Modified	CRC32
..			File folder		
flutter	2,210,674,6...	864,188,383	File folder	28-05-2021 04:...	

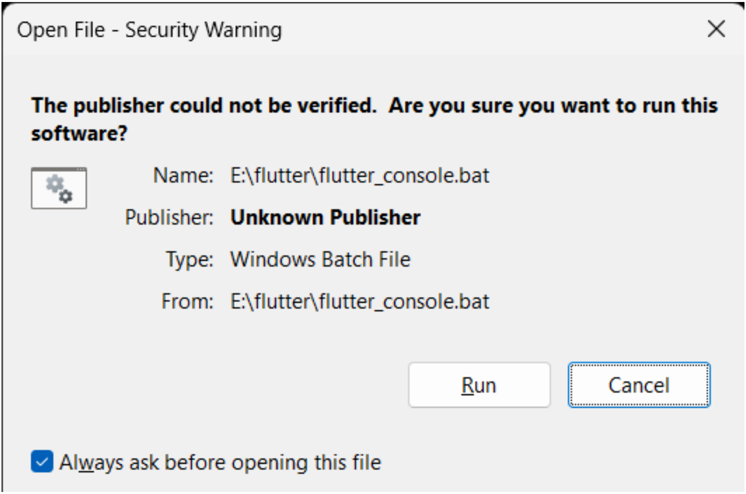
Now double-click on the **'flutter'** folder. Go to **'flutter_console.bat' file** and double-click to open a command prompt window.



engine	05-04-2025 12:27	File folder	
examples	20-12-2024 20:37	File folder	
packages	20-12-2024 20:39	File folder	
.ci.yaml	05-04-2025 12:27	Yaml Source File	211 KB
.gitattributes	20-12-2024 20:29	Git Attributes Sour...	1 KB
.gitignore	05-04-2025 12:27	Git Ignore Source ...	4 KB
analysis_options.yaml	05-04-2025 12:27	Yaml Source File	13 KB



AUTHORS	05-04-2025 12:27	File	6 KB
CHANGELOG.md	05-04-2025 12:27	Markdown Source ...	82 KB
CODE_OF_CONDUCT.md	20-12-2024 20:29	Markdown Source ...	3 KB
CODEOWNERS	05-04-2025 12:27	File	1 KB
CONTRIBUTING.md	05-04-2025 12:27	Markdown Source ...	12 KB
dartdoc_options.yaml	20-12-2024 20:29	Yaml Source File	2 KB
DEPS	05-04-2025 12:27	File	40 KB
flutter_console.bat	20-12-2024 20:29	Windows Batch File	2 KB
flutter_root.iml	20-12-2024 20:29	IML File	1 KB
hs_err_pid25420.log	19-03-2025 18:02	Text Document	14 KB
LICENSE	20-12-2024 20:29	File	2 KB
PATENT_GRANT	20-12-2024 20:29	File	2 KB
README.md	05-04-2025 12:27	Markdown Source ...	7 KB
TESTOWNERS	05-04-2025 12:27	File	28 KB
version	05-04-2025 12:29	File	1 KB



Open File - Security Warning

The publisher could not be verified. Are you sure you want to run this software?

Name: E:\flutter\flutter_console.bat

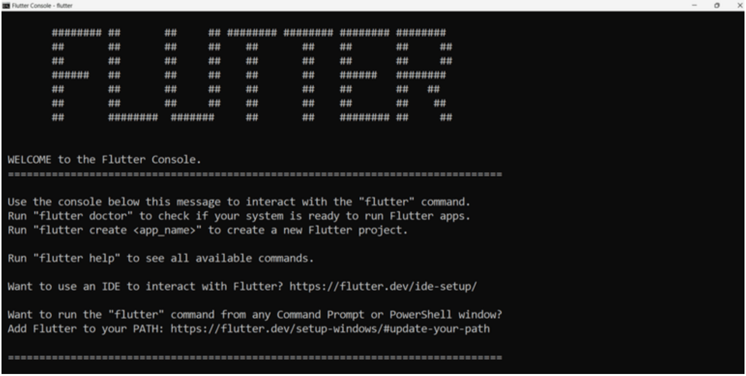
Publisher: **Unknown Publisher**

Type: Windows Batch File

From: E:\flutter\flutter_console.bat

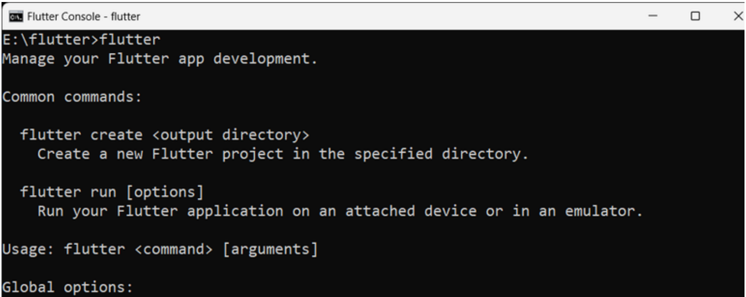
☑ Always ask before opening this file

It should look something like this:

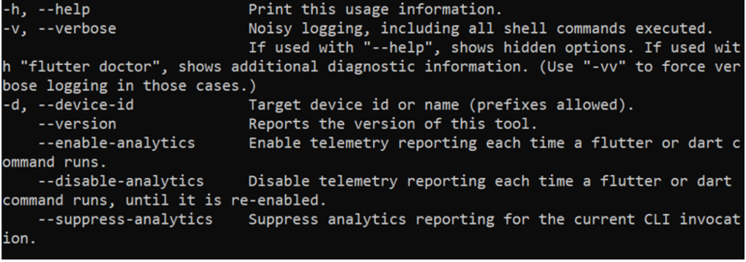


```
 ######## ##       ##     ## ######## ######## ######## ########
 ##       ##       ##     ##    ##       ##    ##       ##     ##
 ##       ##       ##     ##    ##       ##    ##       ##     ##
 ######   ##       ##     ##    ##       ##    ######   ########
 ##       ##       ##     ##    ##       ##    ##       ##   ##
 ##       ##       ##     ##    ##       ##    ##       ##    ##
 ##       ########  #######     ##       ##    ######## ##     ##


WELCOME to the Flutter Console.
===============================================================================

Use the console below this message to interact with the "flutter" command.
Run "flutter doctor" to check if your system is ready to run Flutter apps.
Run "flutter create <app_name>" to create a new Flutter project.

Run "flutter help" to see all available commands.

Want to use an IDE to interact with Flutter? https://flutter.dev/ide-setup/

Want to run the "flutter" command from any Command Prompt or PowerShell window?
Add Flutter to your PATH: https://flutter.dev/setup-windows/#update-your-path

===============================================================================
```

This console is actually a Windows terminal available for the developer to run flutter commands. Type in **'flutter'** to get a list of all the flutter commands that can be run.



```
E:\flutter>flutter
Manage your Flutter app development.

Common commands:

  flutter create <output directory>
    Create a new Flutter project in the specified directory.

  flutter run [options]
    Run your Flutter application on an attached device or in an emulator.

Usage: flutter <command> [arguments]

Global options:
```



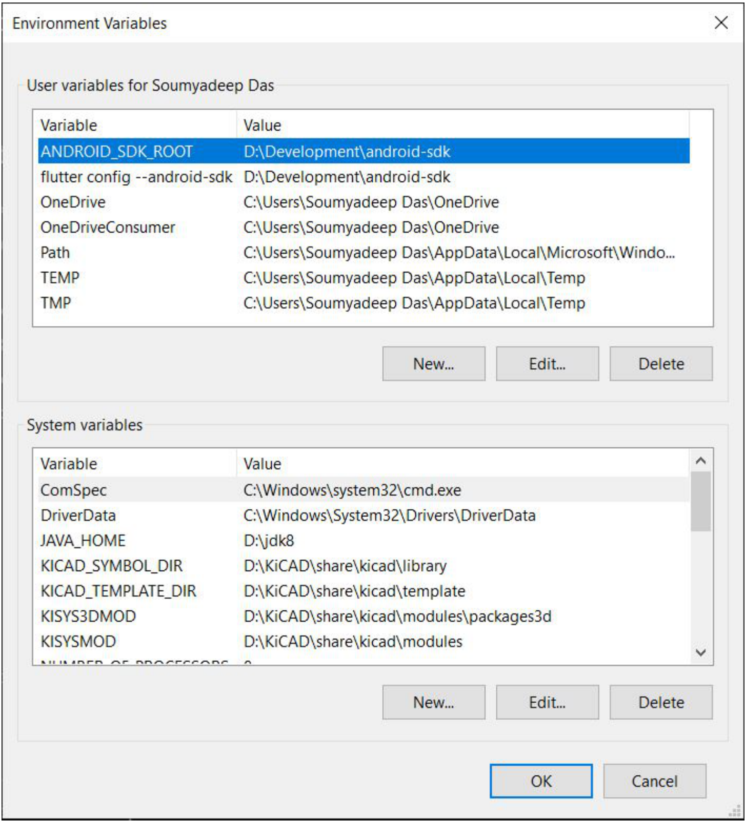
```
-h, --help                  Print this usage information.
-v, --verbose               Noisy logging, including all shell commands executed.
                            If used with "--help", shows hidden options. If used wit
h "flutter doctor", shows additional diagnostic information. (Use "-vv" to force ver
bose logging in those cases.)
-d, --device-id             Target device id or name (prefixes allowed).
    --version               Reports the version of this tool.
    --enable-analytics      Enable telemetry reporting each time a flutter or dart c
ommand runs.
    --disable-analytics     Disable telemetry reporting each time a flutter or dart
command runs, until it is re-enabled.
    --suppress-analytics    Suppress analytics reporting for the current CLI invocat
ion.
```

Whilst it is pretty good to have a terminal to execute flutter commands and create projects, it'd still be better and more convenient to store all our flutter projects somewhere else on our system for easy access. Let us steer over to the next step of our journey!

Step 4: Check and edit environment variables for global system access.

For this, scroll down to **'Update your path'** on the official Docs page of the flutter installation page. For this, go to **Control Panel > System and Security > System > Advanced System Settings > Environment Variables...** .

A dialog box displaying a list of the available environment variables appears on your screen.



Environment Variables

User variables for Soumyadeep Das

Variable	Value
ANDROID_SDK_ROOT	D:\Development\android-sdk
flutter config --android-sdk	D:\Development\android-sdk
OneDrive	C:\Users\Soumyadeep Das\OneDrive
OneDriveConsumer	C:\Users\Soumyadeep Das\OneDrive
Path	C:\Users\Soumyadeep Das\AppData\Local\Microsoft\Windo...
TEMP	C:\Users\Soumyadeep Das\AppData\Local\Temp
TMP	C:\Users\Soumyadeep Das\AppData\Local\Temp

System variables

Variable	Value
ComSpec	C:\Windows\system32\cmd.exe
DriverData	C:\Windows\System32\Drivers\DriverData
JAVA_HOME	D:\jdk8
KICAD_SYMBOL_DIR	D:\KiCAD\share\kicad\library
KICAD_TEMPLATE_DIR	D:\KiCAD\share\kicad\template
KISYS3DMOD	D:\KiCAD\share\kicad\modules\packages3d
KISYSMOD	D:\KiCAD\share\kicad\modules

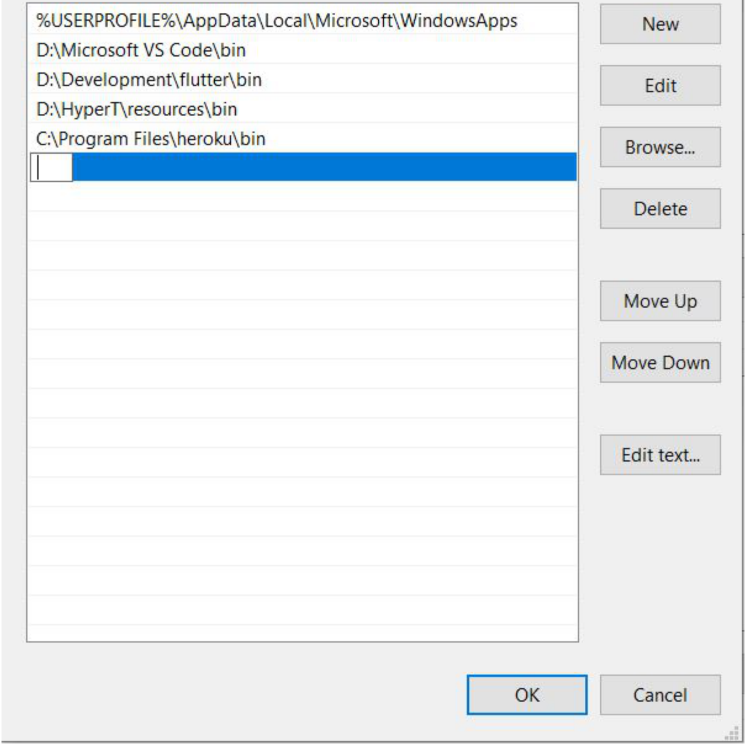
Environment Variables are global system variables present at the root level, which aids in configuring various aspects of Windows.

We will now add the flutter tool as an environment variable for direct access (instead of running the .bat executable), and unlock it on the entire PowerShell and Command Prompt of your system.

To do this, glance through the following steps:

- Check for **'Path'** variable under **User Variables** list. If not already present, create a new variable **('New...')** and assign the '**flutter\bin**' directory as its value.
- Now double-click on the **'Path'** variable and add a new entry by double-clicking on a column below. It should look something like this:

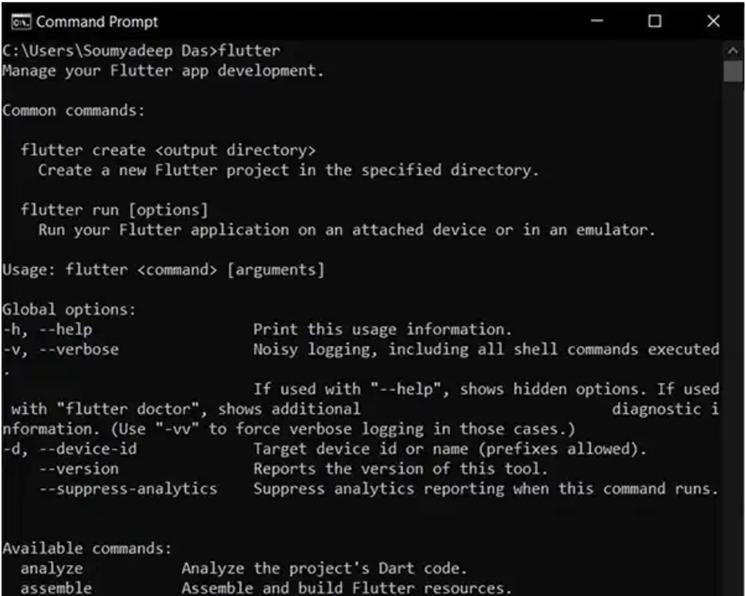
Edit environment variable



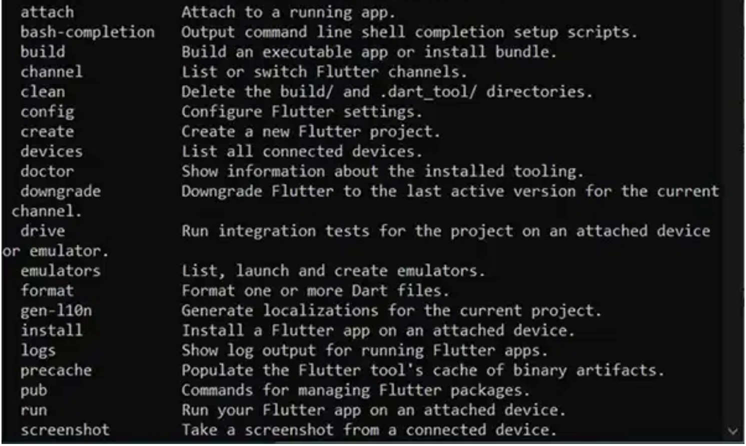
```
%USERPROFILE%\AppData\Local\Microsoft\WindowsApps
D:\Microsoft VS Code\bin
D:\Development\flutter\bin
D:\HyperT\resources\bin
C:\Program Files\heroku\bin
```

- In the path, copy the entire directory of **flutter\bin** folder and paste it. Click **'Ok'** twice to complete the setup. Now, make sure that you have closed any existing Command Prompt/Windows PowerShell windows that are open.

Now, check whether your flutter framework can be accessed globally. To do this, open any terminal (say Command Prompt) and type in '**flutter**' and see whether you get the same list of commands as you did get earlier from the .bat terminal. If yes, you have successfully completed setting up flutter on the root level in your system. *If not, you might as well consider re-running the setup again.*



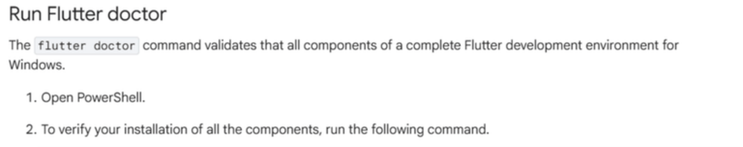
```
C:\Users\Soumyadeep Das>flutter
Manage your Flutter app development.

Common commands:

  flutter create <output directory>
    Create a new Flutter project in the specified directory.

  flutter run [options]
    Run your Flutter application on an attached device or in an emulator.

Usage: flutter <command> [arguments]

Global options:
-h, --help                  Print this usage information.
-v, --verbose               Noisy logging, including all shell commands executed
.
                            If used with "--help", shows hidden options. If used
 with "flutter doctor", shows additional                        diagnostic i
nformation. (Use "-vv" to force verbose logging in those cases.)
-d, --device-id             Target device id or name (prefixes allowed).
    --version               Reports the version of this tool.
    --suppress-analytics    Suppress analytics reporting when this command runs.


Available commands:
  analyze            Analyze the project's Dart code.
  assemble           Assemble and build Flutter resources.
```



```
  attach            Attach to a running app.
  bash-completion   Output command line shell completion setup scripts.
  build             Build an executable app or install bundle.
  channel           List or switch Flutter channels.
  clean             Delete the build/ and .dart_tool/ directories.
  config            Configure Flutter settings.
  create            Create a new Flutter project.
  devices           List all connected devices.
  doctor            Show information about the installed tooling.
  downgrade         Downgrade Flutter to the last active version for the current
 channel.
  drive             Run integration tests for the project on an attached device
or emulator.
  emulators         List, launch and create emulators.
  format            Format one or more Dart files.
  gen-l10n          Generate localizations for the current project.
  install           Install a Flutter app on an attached device.
  logs              Show log output for running Flutter apps.
  precache          Populate the Flutter tool's cache of binary artifacts.
  pub               Commands for managing Flutter packages.
  run               Run your Flutter app on an attached device.
  screenshot        Take a screenshot from a connected device.
```

Step 5: Now, you have to analyze and check whether something is missing/has to be installed further.

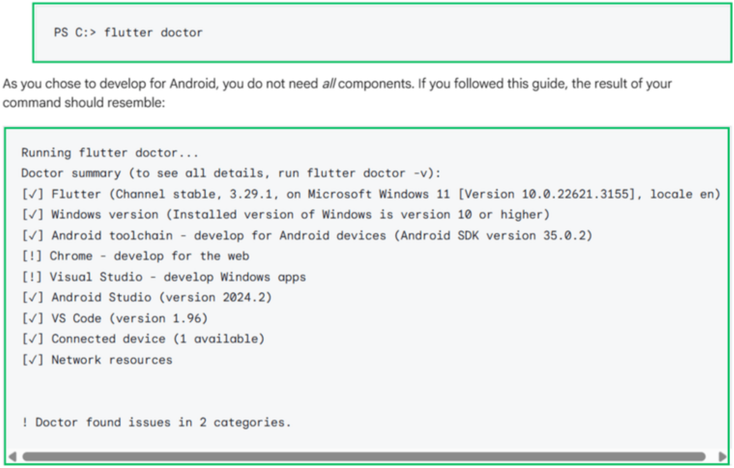
To do this, under the Command Prompt terminal, type in '**Flutter Doctor**' to check for other requirements.



Run Flutter doctor

The `flutter doctor` command validates that all components of a complete Flutter development environment for Windows.

1. Open PowerShell.
2. To verify your installation of all the components, run the following command.

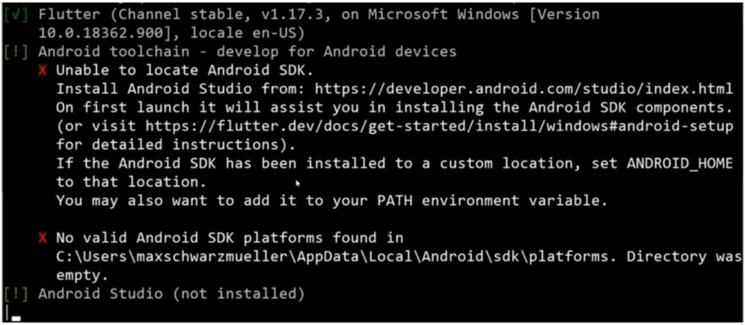


```
PS C:> flutter doctor
```

As you chose to develop for Android, you do not need *all* components. If you followed this guide, the result of your command should resemble:

```
Running flutter doctor...
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 3.29.1, on Microsoft Windows 11 [Version 10.0.22621.3155], locale en)
[✓] Windows version (Installed version of Windows is version 10 or higher)
[✓] Android toolchain - develop for Android devices (Android SDK version 35.0.2)
[!] Chrome - develop for the web
[!] Visual Studio - develop Windows apps
[✓] Android Studio (version 2024.2)
[✓] VS Code (version 1.96)
[✓] Connected device (1 available)
[✓] Network resources


! Doctor found issues in 2 categories.
```

(Since a version has already been installed on my computer, below is an image shown from a previous version, to help you get an understanding of the 'errors' that appear after flutter doctor analysis.)

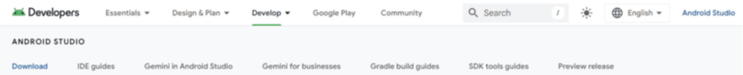


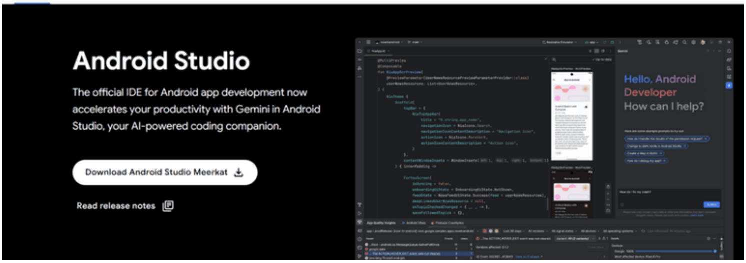
```
[✓] Flutter (Channel stable, v1.17.3, on Microsoft Windows [Version
    10.0.18362.900], locale en-US)
[!] Android toolchain - develop for Android devices
    X Unable to locate Android SDK.
      Install Android Studio from: https://developer.android.com/studio/index.html
      On first launch it will assist you in installing the Android SDK components.
      (or visit https://flutter.dev/docs/get-started/install/windows#android-setup
      for detailed instructions).
      If the Android SDK has been installed to a custom location, set ANDROID_HOME
      to that location.
      You may also want to add it to your PATH environment variable.

    X No valid Android SDK platforms found in
      C:\Users\maxschwarzmueller\AppData\Local\Android\sdk\platforms. Directory was
      empty.
[!] Android Studio (not installed)
```

According to the flutter doctor check, we see that flutter was installed successfully in our system, but the Android tools are missing, and so is Android Studio. We also see that there are no connected devices too. Eventually, the next step is about setting up Android tools on your device, to execute the flutter apps built by you.

Step 6: Setting up Android tools and emulator for android devices.

The first step is to download and install Android Studio. To do this, navigate to the official page of Android Studio and click on '**Download Android Studio**'.

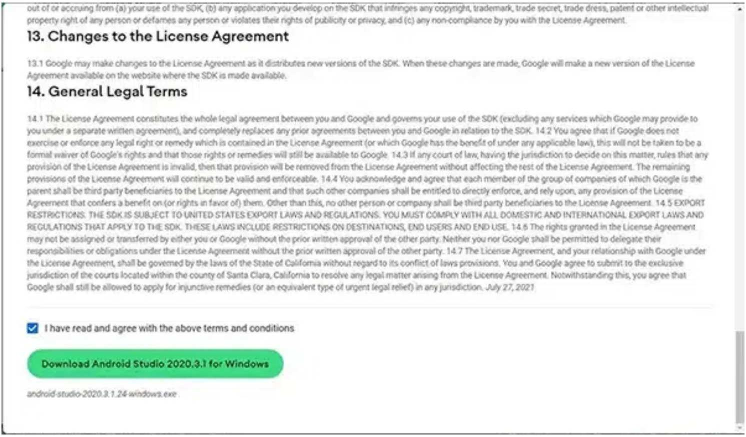




Android Studio

The official IDE for Android app development now accelerates your productivity with Gemini in Android Studio, your AI-powered coding companion.

After accepting the license agreements, you are good to go! Click on the final **Download** button to start downloading.



out of or accruing from (a) your use of the SDK, (b) any application you develop on the SDK that infringes any copyright, trademark, trade secret, trade dress, patent or other intellectual property right of any person or defames any person or violates their rights of publicity or privacy, and (c) any non-compliance by you with the License Agreement.

13. Changes to the License Agreement

13.1 Google may make changes to the License Agreement as it distributes new versions of the SDK. When these changes are made, Google will make a new version of the License Agreement available on the website where the SDK is made available.

14. General Legal Terms

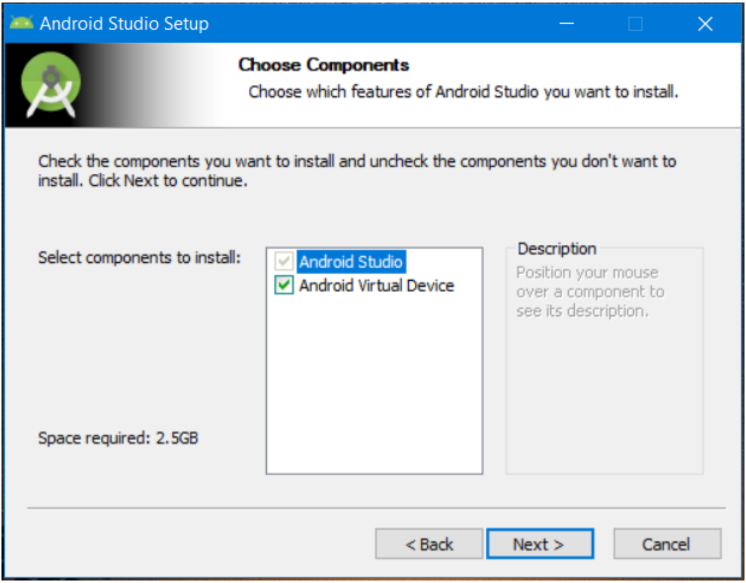
14.1 The License Agreement constitutes the whole legal agreement between you and Google and governs your use of the SDK (excluding any services which Google may provide to you under a separate written agreement), and completely replaces any prior agreements between you and Google in relation to the SDK. 14.2 You agree that if Google does not exercise or enforce any legal right or remedy which is contained in the License Agreement (or which Google has the benefit of under any applicable law), this will not be taken to be a formal waiver of Google's rights and that those rights or remedies will still be available to Google. 14.3 If any court of law, having the jurisdiction to decide on this matter, rules that any provision of the License Agreement is invalid, then that provision will be removed from the License Agreement without affecting the rest of the License Agreement. The remaining provisions of the License Agreement will continue to be valid and enforceable. 14.4 You acknowledge and agree that each member of the group of companies of which Google is the parent shall be third party beneficiaries to the License Agreement and that such other companies shall be entitled to directly enforce, and rely upon, any provision of the License Agreement that confers a benefit on (or rights in favor of) them. Other than this, no other person or company shall be third party beneficiaries to the License Agreement. 14.5 EXPORT RESTRICTIONS. THE SDK IS SUBJECT TO UNITED STATES EXPORT LAWS AND REGULATIONS. YOU MUST COMPLY WITH ALL DOMESTIC AND INTERNATIONAL EXPORT LAWS AND REGULATIONS THAT APPLY TO THE SDK. THESE LAWS INCLUDE RESTRICTIONS ON DESTINATIONS, END USERS AND END USE. 14.6 The rights granted in the License Agreement may not be assigned or transferred by either you or Google without the prior written approval of the other party. Neither you nor Google shall be permitted to delegate their responsibilities or obligations under the License Agreement without the prior written approval of the other party. 14.7 The License Agreement, and your relationship with Google under the License Agreement, shall be governed by the laws of the State of California without regard to its conflict of laws provisions. You and Google agree to submit to the exclusive jurisdiction of the courts located within the county of Santa Clara, California to resolve any legal matter arising from the License Agreement. Notwithstanding this, you agree that Google shall still be allowed to apply for injunctive remedies (or an equivalent type of urgent legal relief) in any jurisdiction. *July 27, 2021*

☑ I have read and agree with the above terms and conditions

android-studio-2020.3.1.24-windows.exe

After the download is complete, let's move on to the next step, i.e. installation.

Under '**Components**', make sure that both Android Studio and Android Virtual Device are checked, and only then proceed. The Android Virtual Device is an essential tool for running various types and sizes of android emulators to test your flutter project. Henceforth, click on '**Next**'.



Choose which features of Android Studio you want to install.

Check the components you want to install and uncheck the components you don't want to install. Click Next to continue.

Select components to install:

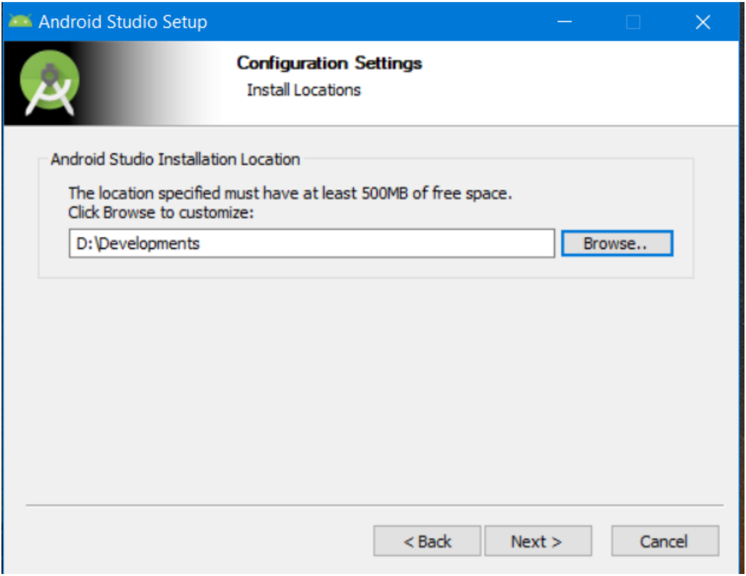
☑ Android Studio
☑ Android Virtual Device

Position your mouse over a component to see its description.

Space required: 2.5GB

Select the directory you would want your file to be installed in. It is recommended to select some other path apart from the system drive.

Once done, click on '**Next**'.



Configuration Settings

Install Locations

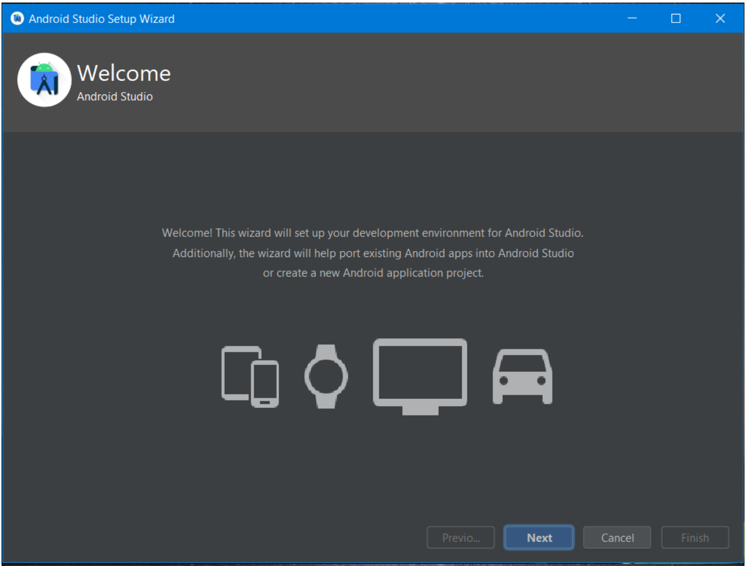
Android Studio Installation Location

The location specified must have at least 500MB of free space.
Click Browse to customize:

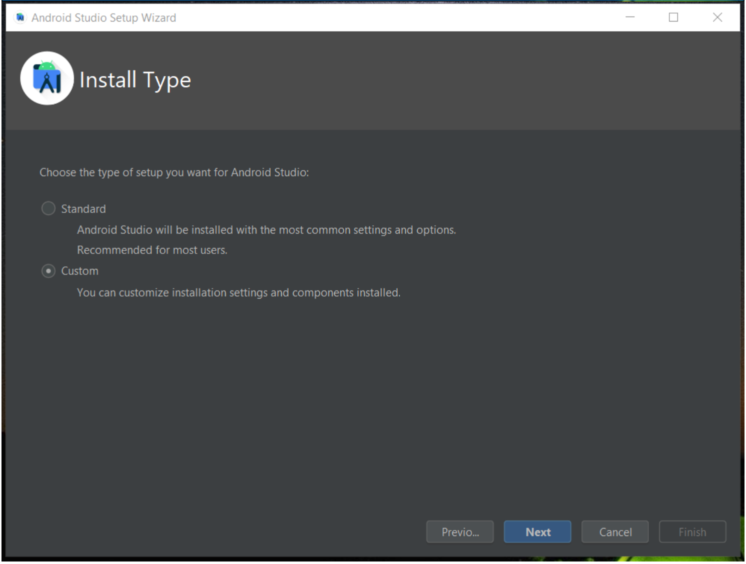
D:\Developments

Finally, click on '**Install**'. Wait for a couple of seconds for the installation to complete. Check the box beside '**Launch Android Studio**'. Click on '**Finish**'.

Wait for Android Studio to launch on your computer. On the home screen, click **Next > Custom > Next**.



Welcome! This wizard will set up your development environment for Android Studio. Additionally, the wizard will help port existing Android apps into Android Studio or create a new Android application project.



Choose the type of setup you want for Android Studio:

○ Standard

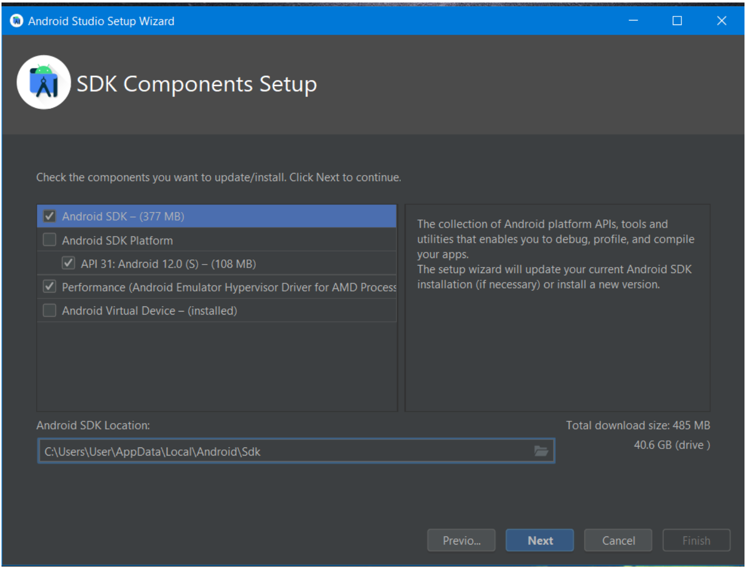
Android Studio will be installed with the most common settings and options.
Recommended for most users.

● Custom

You can customize installation settings and components installed.

For the Java Development kit location in the next step, it is recommended to keep the default path it requires, to avoid the hassle. In the next step, choose the UI appearance you'd like for Android Studio. Click '**Next**'.

This next step is a bit important. Remember to check the required boxes exactly as shown below. If kits have already been installed, you can ignore those and move on. Click '**Next**'. Set your desired folder for Android SDK.



SDK Components Setup

Check the components you want to update/install. Click Next to continue.

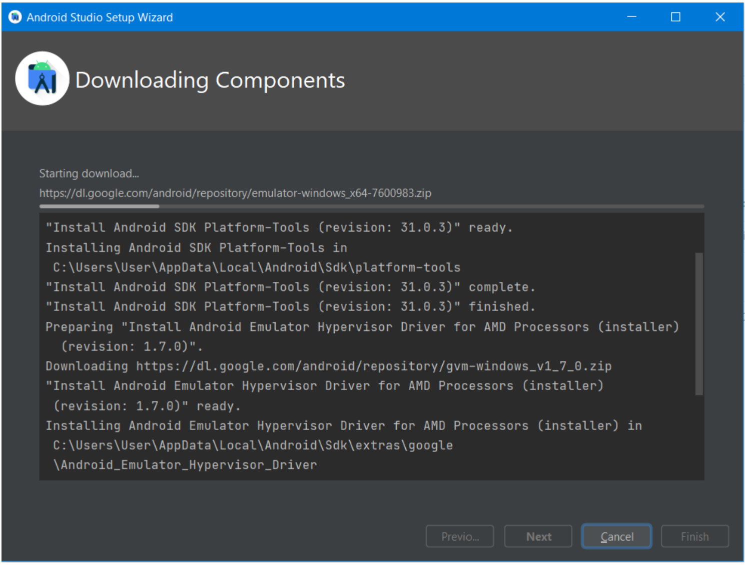
- ☑ Android SDK – (377 MB)
- ☐ Android SDK Platform
  - ☑ API 31: Android 12.0 (S) – (108 MB)
- ☑ Performance (Android Emulator Hypervisor Driver for AMD Process
- ☐ Android Virtual Device – (installed)

The collection of Android platform APIs, tools and utilities that enables you to debug, profile, and compile your apps.
The setup wizard will update your current Android SDK installation (if necessary) or install a new version.

Android SDK Location: C:\Users\User\AppData\Local\Android\Sdk

Total download size: 485 MB
40.6 GB (drive)

With that done, click on '**Finish**'. Android Studio will now install all the necessary android tools required for the execution of your flutter projects. This may take a significant time - it's better to wait!



Starting download...

https://dl.google.com/android/repository/emulator-windows_x64-7600983.zip

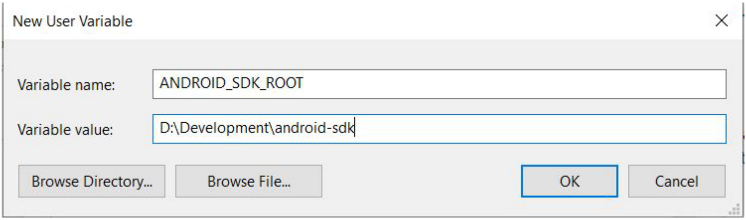
```
"Install Android SDK Platform-Tools (revision: 31.0.3)" ready.
Installing Android SDK Platform-Tools in
 C:\Users\User\AppData\Local\Android\Sdk\platform-tools
"Install Android SDK Platform-Tools (revision: 31.0.3)" complete.
"Install Android SDK Platform-Tools (revision: 31.0.3)" finished.
Preparing "Install Android Emulator Hypervisor Driver for AMD Processors (installer)
  (revision: 1.7.0)".
Downloading https://dl.google.com/android/repository/gvm-windows_v1_7_0.zip
"Install Android Emulator Hypervisor Driver for AMD Processors (installer)
 (revision: 1.7.0)" ready.
Installing Android Emulator Hypervisor Driver for AMD Processors (installer) in
 C:\Users\User\AppData\Local\Android\Sdk\extras\google
 \Android_Emulator_Hypervisor_Driver
```

Now, we are ready to create and build flutter projects on Android Studio and run it on a real or a virtual Android device (emulator).

Step 7: Set SDK as an environment variable, for global access.

Now, open Command Prompt terminal and run 'flutter doctor' again. If you have installed Android SDK in the default directory suggested by Android Studio, there wouldn't be any problem that would appear. Nevertheless, if you have installed it in a non-default directory, flutter would not be able to detect it in your system. To help it able to do that, you guessed it...we would be assigning it as an environment variable, **giving global access**.

As discussed earlier in Step 4, go to environment variables and click '**New**', and do the following (as recommended by flutter doctor). Click '**OK**'.



Variable name: ANDROID_SDK_ROOT

Variable value: D:\Development\android-sdk

Step 8: Accept required Android Licenses.

On the Command Prompt terminal, type in:

```
flutter doctor --android-licenses
```

as suggested by flutter doctor. Hit Enter. To review licenses, type '**y**' for Yes.

You'll see a couple of repeated prompts that look like this:

```
Accept? (y/N):
```

Type '**y**' whenever asked for.

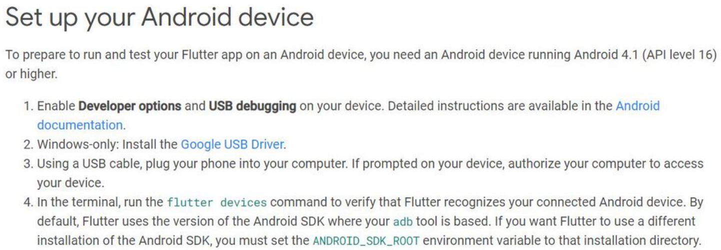
Finally, after all the license agreements have been accepted, you should see a message that looks something like this:

```
All SDK package licenses accepted
```

Step 9: Setup Android Emulator.

You have the option to choose between an Android Device or an Android Emulator to build your application on. It depends totally on you.

For setting up Android Device, go through the official docs page and follow the exact steps as mentioned. Download The Google USB Driver by following the link and install according to the instructions given. This can also be installed through Android Studio, which you can later connect to a real Android Device to build the application.



Set up your Android device

To prepare to run and test your Flutter app on an Android device, you need an Android device running Android 4.1 (API level 16) or higher.

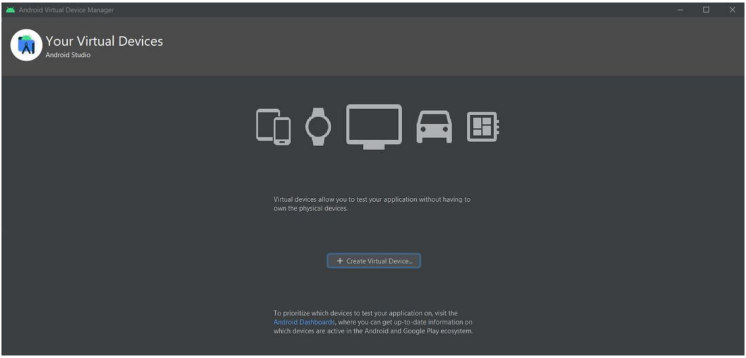
1. Enable **Developer options** and **USB debugging** on your device. Detailed instructions are available in the Android documentation.
2. Windows-only: Install the Google USB Driver.
3. Using a USB cable, plug your phone into your computer. If prompted on your device, authorize your computer to access your device.
4. In the terminal, run the `flutter devices` command to verify that Flutter recognizes your connected Android device. By default, Flutter uses the version of the Android SDK where your `adb` tool is based. If you want Flutter to use a different installation of the Android SDK, you must set the `ANDROID_SDK_ROOT` environment variable to that installation directory.

For setting up Android Emulator, you need to go through the following steps:

- Open **Android Studio.**
- On the topmost menu bar, click on **Tools > SDK Manager.**
- Verify whether you have the latest SDK installed. Remember to install the latest stable version too by checking on the box to the left.

  In my case, it is '**Android 9.0 (Pie)**'. *You can even uncheck the latest version (if not stable), to not only save space but also run all your applications on the stable version itself.*

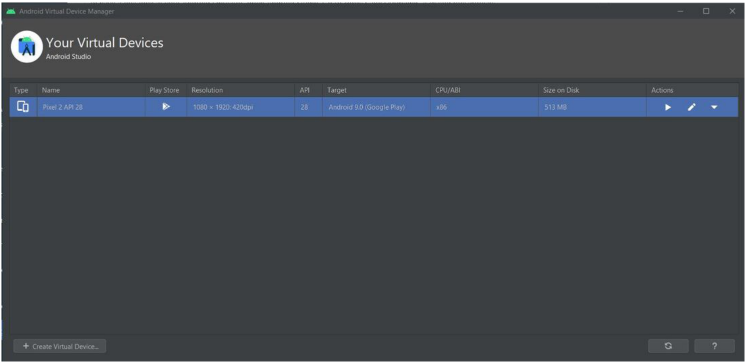
- Under the '**SDK Tools**' tab, don't forget to check Google USB Driver to later connect a real Android Device. With that, click '**Apply**'. Click '**OK**' to start SDK installation. This might take a couple of minutes to complete. After the setup is done, click on '**Finish**'. Your setup is now complete!
- To have a first look at your Android Emulator, open Android Studio. Go to **Tools > AVD Manager**. A dialog box appears.



Virtual devices allow you to test your application without having to own the physical devices.

To prioritize which devices to test your application on, visit the Android Dashboards, where you can get up-to-date information on which devices are active in the Android and Google Play ecosystem.

- Click on '**Create Virtual Device...**', select a device and its dimensions according to your preference, select a system image and lastly, under all default settings, click on 'Finish'. Click on the '▶' button to fire up your emulator.



Type	Name	Play Store	Resolution	API	Target	CPU/ABI	Size on Disk	Actions
	Pixel 2 API 28	▶	1080 × 1920: 420dpi	28	Android 9.0 (Google Play)	x86	513 MB	▶ ✎ ▼

There you go! You now have a fully functional flutter framework with devices/emulators to build your beautiful apps on. Go crazy!

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